

Louie Yin

Waterloo Software Engineer x Varsity Athlete

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EDUCATION

University of Waterloo – Bachelor of Software Engineering September 2024 – Present
• GPA: **3.81/4.0** (equivalent 87%), **Dean's List**, **2x USPORTS Academic All-Canadian**

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, C, C++, Swift, Objective-C, Java, Scala, SQL, HTML/CSS, Assembly
Technologies: React, Next.js, Node.js, Express, FastAPI, Flask, Spring Boot, SwiftUI, Tailwind, Socket.IO
Tools: Docker, Kubernetes, AWS, GCP, Linux/Unix, MySQL, PostgreSQL, MongoDB, Redis, Xcode, Claude Code

EXPERIENCE

Gale (YC W25) January 2026 – Present
Software Engineering Intern San Francisco, CA, US

- Built a custom XFA-to-HTML5 compiler and runtime, saving **\$100,000/year** in Adobe Acrobat licensing fees
- Created an **end-to-end compiler pipeline** that parses and transforms XML into DOM scripts with zero manual field remapping (Python, TypeScript), cutting conversion time to under **20 seconds (10x faster)**
- Built a **JavaScript runtime** to emulate in-browser XFA behaviour with event-driven updates (**500+ fields**)
- Automated PDF prefill by building a **Java Spring Boot** service that maps client questionnaire inputs to PDF XML fields with dynamic font scaling, reducing data entry time by **75%**
- Designed a **Flask + SQLite** persistence layer to enable collaboration across forms with **150ms debounced** API batching that queues all requests into one, **reducing API calls by 97%**
- Built an LLM-orchestrated Slack agent (Kimi K2.5) enabling admin to make **Zod-validated** CRUD calls via **tRPC**

Levanta Labs May 2025 – August 2025
Software Engineer Intern Toronto, ON, CA

- Containerized microservices using **Docker** across 10+ production environments, reducing **deployment time by 50%** and eliminating environment inconsistency issues
- Optimized MySQL database performance by implementing **Redis caching** and server-side filtering on Express.js backends, reducing API response times by **60%** and React frontend load times by **6 seconds**
- Spearheaded the development and deployment of an interactive AI ROI Financial Calculator using Canvas API and Astro, improving the client onboarding process by **boosting engagement by 70%**

PROJECTS

[AI Cooking Assistant](#) | Python, JavaScript, HTML, CSS

- Processed **2+ million data points** using OpenAI Batch API to allow LLaMa fine-tuned LLM to generate recipes, reducing processing time by a **factor of 40x** and processing costs by **\$200**
- Used Flask and Websocket to build a **full-stack, voice-enabled assistant**, while integrating TTS/STT via RegEx for real-time assistant-user interaction
- Developed a **custom Raspberry Pi enclosure** using CAD, incorporating temperature and gas sensors to process cooking data while enhancing user safety

[MindView \(DeltaHacks 2026 Finalist\)](#) | Next.js, Three.js, FastAPI, MongoDB, Gemini API, SciPy, MONAI

- Developed an interactive 3D brain viewer generated from MRI scans to help physicians visualize brain tumours
- Implemented a timeline viewer by interpolating voxel volumes across scans to visualize tumour growth patterns
- Built a **Gemini** chatbot to summarize patient data and analyze brain voxel volumes for tumor growth patterns

[MacNewFile \(3000+ Users\)](#) | Objective-C, Xcode, Open-Source, Greptile

- Built a MacOS Finder Extension that extends the right-click menu functionality, earning **70+ GitHub stars**
- Managed **open-source** contributions by reviewing PRs (via Greptile) and validating changes through local testing

[Terence](#) | Python, GitHub API, PyTest, Twine, PyPI, CI/CD pipelines

- Published a Python package that abstracts GitHub API interactions into a single class to simplify data retrieval

HONOURS AND CERTIFICATIONS

- Governor General's Medal of Canada** – Awarded for highest academic average in school
- Paris 2024 Olympic Swimming Trials Qualifier**, Canadian Nationals Finalist
- RCM Piano Level 10** – First-Class Honours